Module Interaction Diagram

Unified Funeral Services Management Platform

Presentation Layer

- Web Portal
- Mobile App
- Family Portal
- Partner Portal
- API Gateway

API Management Layer

- REST APIs
- GraphQL Endpoints
- WebSocket (Real-time)
- Webhooks
- Rate Limiting

Business Logic Layer

Core Operations Services

- Case Mgmt Service — FSD-01
- Member Registration — FSD-02
- Chain of Custody — FSD-03
- Service & Ceremony — FSD-04
- Payment & Financial — FSD-05
- Inventory & Merch — FSD-06
- Fleet & Equipment — FSD-26
- Staff & Task Mgmt

Customer Experience Services

- Family Portal — FSD-07
- AI Obituary & Memorial — FSD-08
- Handwritten Doc Scanner — FSD-09
- AI Consultation Chatbot — FSD-10

Data Governance Services

- Success Metrics — FSD-11
- RBAC — FSD-12
- Data Dictionary — FSD-13
- Data Lifecycle — FSD-14
- Data Ingestion — FSD-15
- Data Quality — FSD-16
- Data Governance — FSD-17
- Metadata & Ontology — FSD-18
- Data Lineage
- Master Data Mgmt
- Reference Data M
- Privacy & Consent

Analytics Services

- Visitor Analytics — FSD-19
- Data Observability — FSD-20
- Analytics Platform — FSD-21
- KPI Framework — FSD-22

Compliance & Security Services

- Multi-Jurisdiction Compliance — FSD-23
- Cultural & Religious Config — FSD-24
- Audit Trail & Logging — FSD-27
- Data Encryption & Privacy — FSD-28

Integration Services

- Third-Party Integration Hub — FSD-25
- External Data Interface — FSD-31

CMMI Process Services

- Continuous Process Improvement — FSD-29
- Quantitative Project Mgmt — FSD-30

Data Access Layer

- ORM / Repository
- Caching (Redis)
- Search (Elastic)
- Message Queue
- Event Bus

Infrastructure Layer

- Cloud (AWS / Azure)
- Container Orchestration
- Service Mesh
- Monitoring & Logging
- CI/CD Pipeline